

Domáci úkol 1.1

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Trvzení 8.32 (polarizační identity) říká, že v. pr V se sk. s platí:

$$1) \operatorname{Re} \langle u, v \rangle = \frac{1}{2} (\|u+v\|^2 - \|u\|^2 - \|v\|^2)$$

$$2) \operatorname{Im} \langle u, v \rangle = \frac{1}{2} (\|u-iv\|^2 - \|u\|^2 - \|v\|^2)$$

Máme dokázat, že: 1) $\operatorname{Re} \langle u, v \rangle = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$

$$2) \operatorname{Im} \langle u, v \rangle = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

Snaží se dokázat: 1) $\frac{1}{2} (\|u-iv\|^2 - \|u\|^2 - \|v\|^2) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$

$$2) \frac{1}{2} (\|u+v\|^2 - \|u\|^2 - \|v\|^2) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

důkaz:

$$1) 2\|u-iv\|^2 - 2\|u\|^2 - 2\|v\|^2 = \|u-iv\|^2 - \|u+iv\|^2$$

$$\langle u-iv, u-iv \rangle - 2\langle u, u \rangle - 2\langle v, v \rangle + \langle u+iv, u+iv \rangle = 0$$

$$\cancel{\langle u, u \rangle} - \cancel{\langle iv, u \rangle} - \cancel{\langle u, iv \rangle} + \langle iv, iv \rangle - \cancel{2\langle u, u \rangle} - 2\langle v, v \rangle + \cancel{\langle u, u \rangle} + \cancel{\langle iv, u \rangle} + \cancel{\langle u, iv \rangle} + \langle iv, iv \rangle = 0$$

$$2\langle iv, iv \rangle - 2\langle v, v \rangle = 0 \Rightarrow 2(-i)\langle v, v \rangle - 2\langle v, v \rangle = 0$$

$$2) 2(\|u+iv\|^2 - \|u\|^2 - \|v\|^2) = \|u+iv\|^2 - \|u-iv\|^2$$

$$2\langle u+iv, u+iv \rangle - 2\langle u, u \rangle - 2\langle v, v \rangle + \langle u-iv, u-iv \rangle = 0$$

$$\cancel{\langle u, u \rangle} + \cancel{\langle iv, u \rangle} + \cancel{\langle u, iv \rangle} + \cancel{\langle iv, iv \rangle} - \cancel{2\langle u, u \rangle} - 2\langle v, v \rangle + \cancel{\langle u, u \rangle} - \cancel{\langle iv, u \rangle} - \cancel{\langle u, iv \rangle} + \cancel{\langle iv, iv \rangle} = 0 \quad \square$$